

AAD missiles are endo-atmospheric interceptors with capability to intercept targets at a height of 30km, whereas PAD missiles are exo-atmospheric interceptors with a dedicated second stage kill vehicle for interception up to an altitude of 80km. Both the missiles mostly have locally developed components.

DRDO has also developed deadly anti-satellite (ASAT) weapons. Agni-V, with the 'kill vehicle' using ISRO's indigenous Satellite Launch Vehicle (SLV) 3's solid fuel engine and advanced seekers, has the capability to home into hostile satellites. The 15-year 'Technology Perspective and Roadmap' prepared in 2010, has a thrust area for development of ASAT weapons for electronic or physical destruction of satellites in low earth orbit as well as higher geosynchronous orbit to protect space assets.

As part of Make in India and *Atmanirbhar Bharat* initiative, DRDO under Development cum Production Partner (DCPP) programme is committed to handholding of private sector industries to improve their development and production cycle of complex defence systems.

Defence Space Authority and Defence Space Research Agency

The space warfare is not limited to using offensive capabilities in space; it is intrinsically linked with combat on the ground, sea, and air—mak-



Fig. 5: GSAT-7A, an advanced military communications Indian satellite (Source: <https://en.wikipedia.org>)

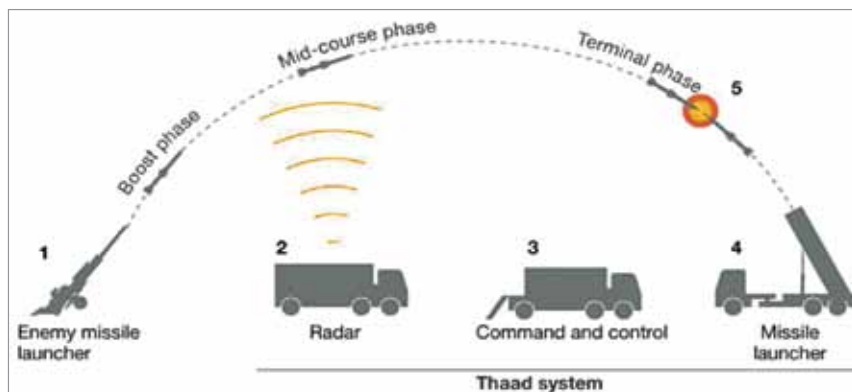


Fig. 6: How the Terminal High Altitude Area Defense (THAAD) system works (Source: Federation of American Scientists, US Department of Defense)

ing it critical for the armed forces to become interoperable with the space force in a conflict situation. Operating in space, unlike in any other domain, involves three dimensions spanning across the globe with no territorial borders. A clearly defined authority functioning as an independent body in the national security architecture is an absolute necessity to achieve the strategic objectives and develop a joint space warfare doctrine.

The Naresh Chandra Task Force, set up in July 2011 to review the recommendations of the Kargil Review Committee and assess the implementation progress and further suggest reforms related to national security, recommended the creation of three new tri-service commands—cyber command, aerospace command, and a special operations command. Accordingly, the Defence Space Agency (DSA), the Defence Cyber Agency (DCA), and the Armed Forces Special Operations Division (AFSOD) were set up in September 2018.

DSA, located in Bengaluru, is a tri-service agency working under the Integrated Defence Staff. It is responsible for operating systems to protect India's interests in outer space and deal with potential space wars

as well as develop a space warfare doctrine and strategy. The Defence Imagery Processing and Analysis Centre, Delhi and the Defence Satellite Control Centre, Bhopal were included in the DSA.

To provide the technical and R&D support, a new agency—the Defence Space Research Agency (DSRO)—was created with the task of developing space warfare weapon systems and technologies to fight wars in space. The DSA is likely to be converted into a full-size tri-service military command in future.

India's anti-missile defence programme

China demonstrated the capability of its anti-satellite program (ASAT) to target assets in outer space when it successfully destroyed an orbiting satellite by launching a ground based kinetic kill vehicle. China's growing military might in space had the greatest impact on India's mindset and necessitated it to build its own capability to respond to any potential threat in space.

DRDO began working on developing a multi-layered and multi-phased missile defence system, like the American anti-ballistic missile defence system Terminal High Altitude Area Defence (THAAD), which has the capability to shoot down short, medium, and intermediate-range ballistic missiles in their terminal phase. Under the DRDO's